

Computational Plasma Exploration Project

Boron Impurity Transport in the HSX Stellarator: A 1D Sensitivity Study

Ikhlas Addow

May 1, 2026

Contents

1	Introduction and Motivation	2
2	Background	2
2.1	HSX and Impurity Transport	2
2.2	The Transport Equation	3
2.3	Coronal Equilibrium	3
3	Model Description	3
3.1	Geometry and Grid	3
3.2	Boundary Conditions	4
3.3	Numerical Solver	4
3.4	Input Profiles	4
3.5	Atomic Physics	4
3.6	Neutral Boron Source	5
3.7	Transport Coefficients	5
4	Results	5
4.1	Background Profiles (Figure 1)	5
4.2	OpenADAS Atomic Rates (Figure 2)	6
4.3	Transport Coefficients (Figure 3)	7
4.4	Damköhler Timescale Map (Figure 4)	8
4.5	Charge-State Profiles and Coronal Equilibrium (Figure 5)	9
4.6	Core Penetration and Peaking Factor (Figure 6)	10
4.7	Sensitivity to Anomalous Diffusivity (Figures 7 and 8)	10
5	Discussion	11
5.1	What the Model Gets Right	11
5.2	Limitations and Assumptions	13
5.3	Connection to Course Concepts	13
6	Conclusions and Future Work	14

1 Introduction and Motivation

Boronization is a standard wall-conditioning technique used in magnetic confinement fusion devices. A thin boron film deposited on plasma-facing components reduces oxygen and carbon impurity levels, lowers the effective charge Z_{eff} , and can improve plasma performance significantly. The Helically Symmetric eXperiment (HSX) at the University of Wisconsin Madison has recently been exploring boron powder injection as a real-time wall conditioning method, and understanding where injected boron ends up after ionization, specifically whether it stays near the edge or penetrates into the core, is a practically relevant question.

This project builds a standalone 1D radial impurity transport model for boron in HSX and uses it to study how sensitive the core penetration of boron is to uncertain transport parameters. The model is inspired by the STRAHL/pySTRAHL impurity transport workflow used in tokamak research, adapted to HSX geometry. This is not meant to be a final quantitative prediction. The goal is to build a clear framework, test how the result changes when transport assumptions are varied, and identify which measurements are needed next.

The physics questions driving this project are:

1. How quickly do injected boron neutrals ionize under HSX core conditions, and how far do they penetrate before becoming fully ionized?
2. How sensitive is the steady-state charge-state distribution and core boron fraction to anomalous radial diffusion and an assumed inward pinch?
3. Do the computed charge-state populations approach local coronal equilibrium in the core, and what does that tell us about the relative importance of transport versus atomic physics?

2 Background

2.1 HSX and Impurity Transport

HSX is a quasi-helically symmetric (QHS) stellarator with minor radius $a = 0.12$ m and major radius $R = 1.20$ m. The quasi-helical symmetry suppresses neoclassical transport compared to conventional stellarators, making HSX a useful testbed for distinguishing neoclassical from anomalous transport contributions.

Impurity transport in fusion plasmas is governed by diffusion and convection in the radial direction, coupled to atomic physics processes (ionization and recombination) that continuously redistribute particles among charge states. For an impurity species like boron (nuclear charge $Z = 5$), the charge-state distribution at any radial location depends on the competition between the local ionization and recombination timescales and the radial transport timescale. If transport is slow relative to ionization, the local charge-state distribution approaches the so-called *coronal equilibrium*, where ionization and recombination balance locally without any transport. If transport is fast, the distribution can be “frozen” into a state far from coronal equilibrium.

2.2 The Transport Equation

For each boron charge state Z (with $Z = 0$ for neutral boron through $Z = 5$ for fully stripped B^{5+}), the governing equation in 1D cylindrical geometry is:

$$\frac{\partial n_Z}{\partial t} = -\frac{1}{r} \frac{\partial}{\partial r} (r \Gamma_Z) + \mathcal{I}_Z + \mathcal{R}_Z + S_Z, \quad (1)$$

where r is the minor radius, Γ_Z is the radial particle flux, $\mathcal{I}_Z$ and $\mathcal{R}_Z$ are the ionization and recombination coupling terms, and S_Z is a source term (nonzero only for $Z = 0$).

The radial flux takes the standard diffusion-convection form,

$$\Gamma_Z = -D_Z(r) \frac{\partial n_Z}{\partial r} + V_Z(r) n_Z, \quad (2)$$

where D_Z is the radial diffusivity and V_Z is the convective (pinch) velocity. By convention, $V < 0$ corresponds to inward (core-directed) convection.

The ionization coupling terms are:

$$\mathcal{I}_Z = -S_Z^{\text{ion}}(r) n_Z + S_{Z-1}^{\text{ion}}(r) n_{Z-1}, \quad (3)$$

$$\mathcal{R}_Z = -\alpha_Z^{\text{rec}}(r) n_Z + \alpha_{Z+1}^{\text{rec}}(r) n_{Z+1}, \quad (4)$$

where S_Z^{ion} and α_Z^{rec} are the electron-impact ionization and recombination rate coefficients (in units of s^{-1} , already multiplied by n_e) obtained from the OpenADAS database. Every particle removed from one charge state by ionization or recombination is exactly added to the adjacent state, so the coupling is strictly particle-conservative.

2.3 Coronal Equilibrium

In the limit where transport is negligible, the charge-state fractions $f_Z \equiv n_Z / \sum_Z n_Z$ satisfy the coronal equilibrium condition:

$$\frac{f_{Z+1}}{f_Z} = \frac{S_Z^{\text{ion}}}{\alpha_{Z+1}^{\text{rec}}}. \quad (5)$$

The model computes coronal equilibrium fractions from Eq. (5) at each radial location using the local T_e and n_e , and overlays these on the time-dependent simulation results as a consistency check. Agreement between the simulation and coronal equilibrium in the core indicates that atomic physics has converged self-consistently and that transport is not strongly distorting the ionization balance.

3 Model Description

3.1 Geometry and Grid

The model uses a 1D minor-radius grid with 100 uniformly spaced points from $r = 0$ to $r = a$. The finite-volume discretization is built in cylindrical geometry, with cell volumes

$$\Delta V_i = \int_{r_{i-1/2}}^{r_{i+1/2}} r dr = \frac{1}{2} (r_{i+1/2}^2 - r_{i-1/2}^2), \quad (6)$$

where $r_{i\pm 1/2}$ are the cell-face radii. The toroidal factor $4\pi^2 R$ cancels identically in the divergence operator and is not carried through the numerics. Face areas are simply $A_{\text{face}} = r_{\text{face}}$.

3.2 Boundary Conditions

At $r = 0$, the symmetry (zero-flux) boundary condition is enforced by simply excluding the innermost face from the transport loop. At the last closed flux surface (LCFS), a zero-gradient (Neumann) boundary condition is applied to diffusion (i.e., $\partial n / \partial r = 0$ at $r = a$), and convective outflow is allowed only when $V(a) > 0$. Inflow from outside the LCFS is blocked. The Neumann outer boundary is a standard first-pass choice but can produce mild artificial density accumulation near the LCFS at early times; this is a known limitation of the approach.

3.3 Numerical Solver

The coupled system of six charge-state equations is time-integrated using the BDF (Backward Differentiation Formula) method from `scipy.integrate.solve_ivp`. BDF is an L-stable implicit method and is the correct choice for this problem due to extreme stiffness: the ionization timescale for neutral boron is approximately $\tau_{B^0 \rightarrow B^{1+}} \approx 2 \mu\text{s}$ at HSX core conditions ($T_e \sim 500 \text{ eV}$, $n_e \sim 5 \times 10^{18} \text{ m}^{-3}$), while the simulation runs for $t_{\text{end}} = 50 \text{ ms}$, giving stiffness ratios of order 10^4 .

Solver tolerances are `rtol` = 10^{-5} and `atol` = 10^{-3} m^{-3} . The absolute tolerance floor is many orders of magnitude below expected boron densities (10^{15} to 10^{18} m^{-3}) and is essentially irrelevant in practice.

Convection is discretized with first-order upwinding, which is monotone (no spurious oscillations) but introduces numerical diffusion $D_{\text{num}} \approx |V| \Delta r / 2$. For the grid and transport cases used here, $D_{\text{num}} \lesssim 3 \times 10^{-5} \text{ m}^2/\text{s}$, which is well below the physical diffusivities in all cases.

3.4 Input Profiles

The electron temperature and density profiles are taken from HSX Thomson scattering data for shot 37, averaged over $t = 0.83$ to 0.85 s . The Thomson system provides 10 chords covering $\rho = r/a \in [0.016, 0.906]$, where the innermost chord at $\rho = 0.016$ is essentially on-axis. Profiles are extended to the LCFS using a monotone-preserving PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) interpolant with axis and edge anchors. The on-axis values are pinned to the innermost Thomson chord. The LCFS anchor uses assumed values $T_e = 40 \text{ eV}$ and $n_e = 10^{18} \text{ m}^{-3}$, which are order-of-magnitude estimates for the HSX scrape-off layer and are not measurements.

Ion temperature T_i is taken as a spatially uniform 70 eV , which is a reasonable first guess for HSX but is not constrained by any measurement in this version of the model. Charge-exchange recombination, which would require T_i , is not included.

3.5 Atomic Physics

Boron ionization and recombination rates are obtained from the OpenADAS ADF11 database, specifically the SCD89 (ionization) and ACD89 (recombination) files for boron. These give rate coefficients in units of $\text{cm}^3 \text{ s}^{-1}$ as a function of T_e and n_e . The rate coefficients are interpolated in log-log space using `scipy's RegularGridInterpolator`, and converted to rates in s^{-1} via $S_Z^{\text{ion}} = \langle \sigma v \rangle_{\text{SCD}} \times n_e$. The unit conversion $n_e [\text{cm}^{-3}] = n_e [\text{m}^{-3}] / 10^6$ is applied before the interpolation. Rates are evaluated along the HSX profile using the measured $T_e(r)$ and $n_e(r)$.

3.6 Neutral Boron Source

The source of neutral boron (applied only to the $Z = 0$ equation) is modeled as a Gaussian in normalized radius centered at $\rho = 0.98$ with width $\sigma = 0.035$ in ρ , representing powder entering near the edge. The source is normalized so that

$$\int S_{B^0}(r) dV = \dot{N}, \quad (7)$$

where $\dot{N} = 10^{20} \text{ s}^{-1}$ is the total boron injection rate. This value is an order-of-magnitude placeholder; it has not been calibrated against dropper measurements or boron spectroscopy and should not be used for quantitative comparisons.

3.7 Transport Coefficients

Since boron-specific neoclassical transport coefficients from SFINCS or PENTA are not yet available for HSX, the model uses three generic anomalous diffusivity profiles as sensitivity cases:

$$D(r) = D_{\text{core}} + (D_{\text{edge}} - D_{\text{core}})\rho^2, \quad (8)$$

with parameter values given in Table 1.

Table 1: Anomalous diffusivity cases used in the sensitivity scan. All values in m^2/s . None of these cases is claimed to be the correct answer; they span a plausible range for HSX conditions.

Case	D_{core}	D_{edge}	Notes
Low	0.01	0.05	Near-neoclassical
Medium	0.05	0.40	Moderate anomalous
High	0.15	1.00	Strong anomalous

The inward convective (pinch) velocity in the `charge_scaled` case is:

$$V_Z(r) = -0.08 \frac{Z}{5} \sin^{3/2}(\pi\rho) (0.4 + 0.6\rho) \quad [\text{m/s}], \quad (9)$$

where the denominator 5 normalizes to $Z_{\text{max}} = 5$ (fully stripped boron), so that B^{5+} receives the full amplitude of -0.08 m/s and lower charge states receive proportionally less. This is not derived from any electric field measurement; it is an assumed profile included to test the sensitivity to an inward pinch.

4 Results

4.1 Background Profiles (Figure 1)

Figure 1 shows the plasma profiles used as input to the transport model (`01_profiles_used.png`). The top two panels show the Thomson-measured electron temperature and density, with raw data points, error bars, and the PCHIP interpolating curve. The innermost chord is at $\rho = 0.016$ with $T_e \approx 982 \text{ eV}$, and the outermost chord reaches $\rho = 0.906$ before the profile transitions to the assumed edge anchor. The grey shaded region beyond $\rho \approx 0.906$ marks where the profile is no longer constrained by data. The bottom panel shows the assumed flat ion temperature $T_i = 70 \text{ eV}$.

The steep T_e profile ($\sim 980 \text{ eV}$ on axis, falling to 40 eV at the LCFS) has an important consequence: the ionization rates for boron vary by orders of magnitude between the core and the edge, as shown in the next figure.

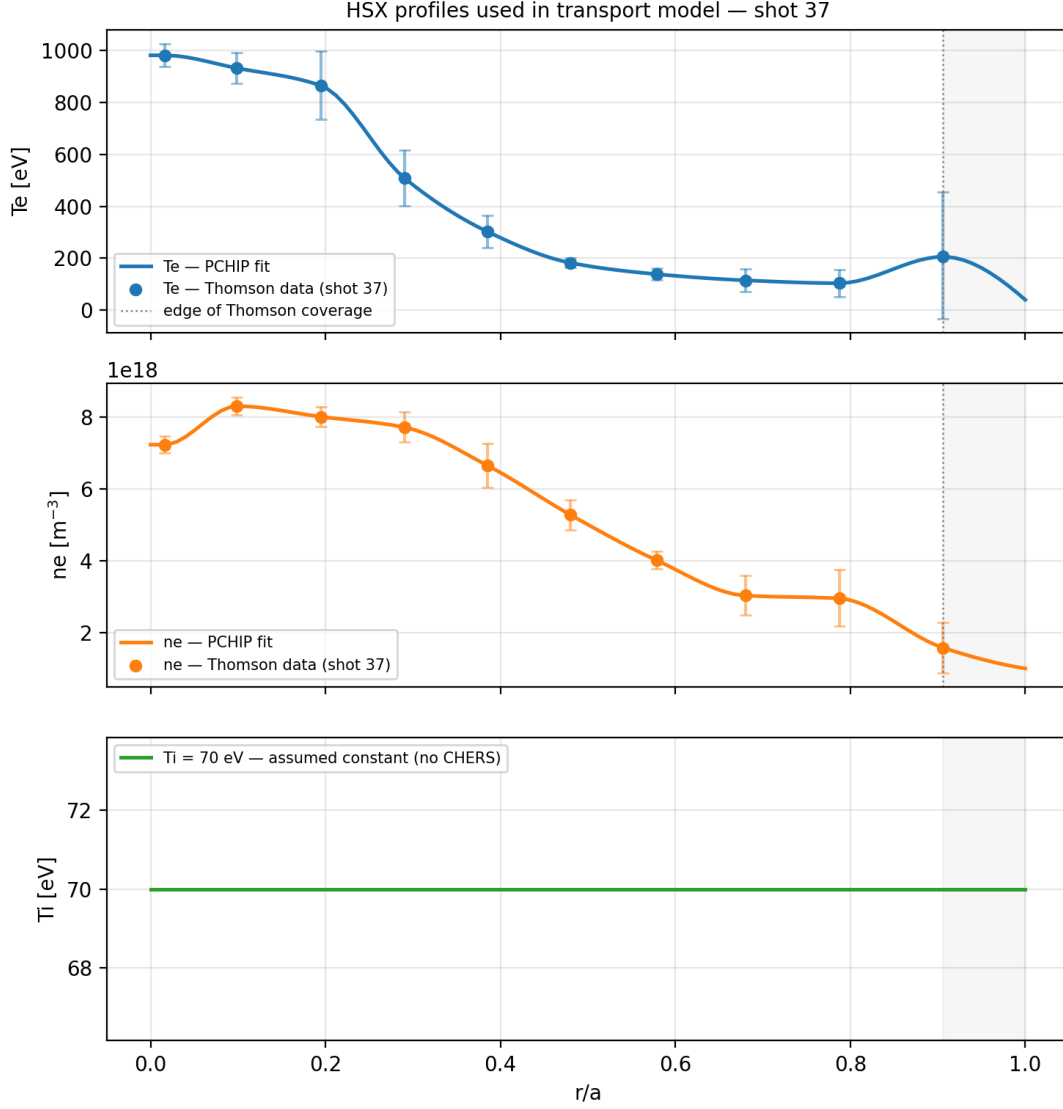


Figure 1: (01_profiles_used.png) Plasma profiles used in the transport model. *Top:* Electron temperature $T_e(\rho)$ from HSX Thomson scattering, shot 37 ($t = 0.83$ to 0.85 s), with raw data points, error bars, and PCHIP interpolating curve. The grey band marks the region beyond Thomson coverage where the profile is assumed. *Middle:* Electron density $n_e(\rho)$, same source. *Bottom:* Ion temperature $T_i = 70$ eV, assumed constant (no CHERS measurement available).

4.2 OpenADAS Atomic Rates (Figure 2)

Figure 2 shows the boron electron-impact ionization and recombination rates evaluated along the HSX profile (02_adas_rates_vs_radius.png). The ionization rate for neutral boron ($B^0 \rightarrow B^{1+}$) reaches $\sim 6 \times 10^5 \text{ s}^{-1}$ in the core and remains high across most of the profile, giving an ionization timescale of ~ 1 to $2 \mu\text{s}$. This is extremely fast compared to any transport timescale, which immediately tells us that a boron neutral injected at the edge will ionize to B^{1+} before it can diffuse more than a tiny fraction of the minor radius. The question of core penetration is therefore not about neutrals reaching the core, it is about how quickly the *ions* B^{1+} through B^{5+} are transported inward.

Higher charge states ionize more slowly: $B^{4+} \rightarrow B^{5+}$ has a rate of $\sim 10^2$ to 10^3 s^{-1} , giving timescales of 1 to 10 ms. This is comparable to the ion transport time, meaning the charge-state distribution in the core is genuinely set by the competition between transport and atomic physics.

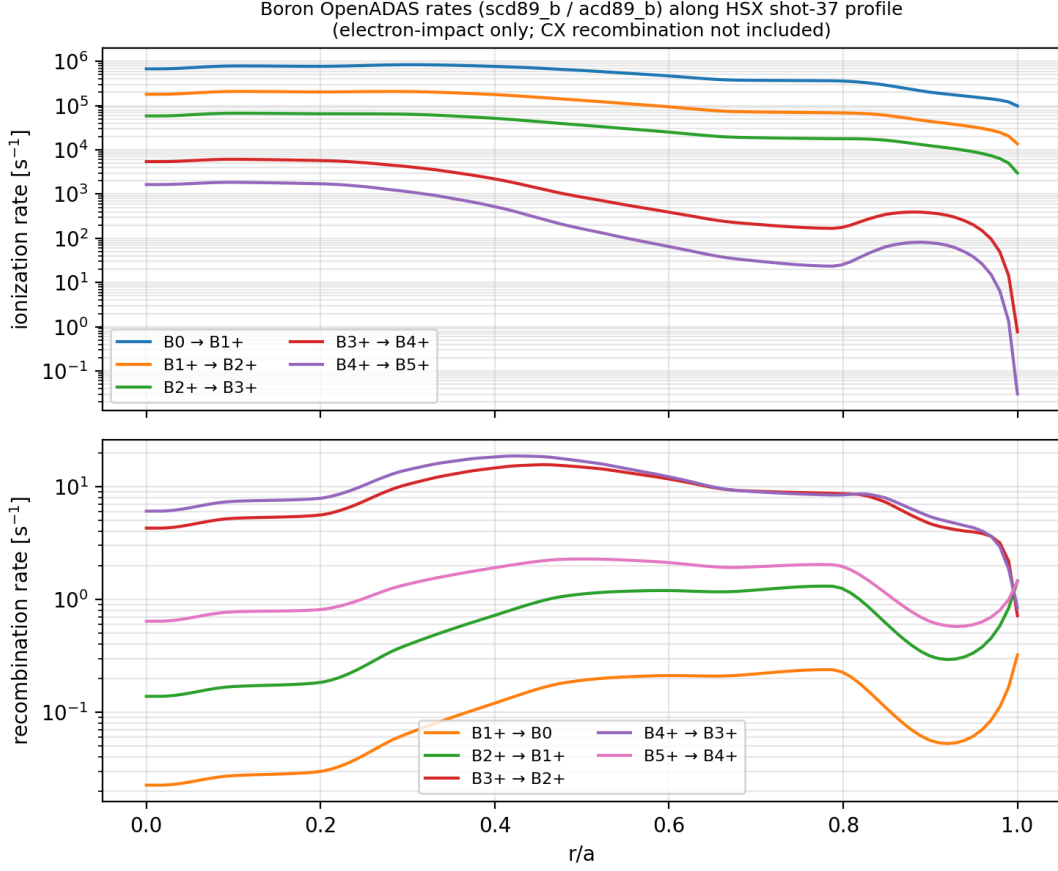


Figure 2: (02_adas_rates_vs_radius.png) Boron electron-impact ionization (top) and recombination (bottom) rates in s^{-1} , evaluated along the measured HSX T_e/n_e profile using OpenADAS SCD89/ACD89 data. The $B^0 \rightarrow B^{1+}$ ionization rate exceeds $5 \times 10^5 \text{ s}^{-1}$ across most of the profile, giving a neutral ionization timescale of order 1 to 2 μs .

4.3 Transport Coefficients (Figure 3)

Figure 3 shows the diffusivity and pinch velocity profiles used in the medium-D, charge-scaled-pinch reference case (03_transport_coefficients_used.png). The diffusivity rises from $D_{\text{core}} = 0.05 \text{ m}^2/\text{s}$ on axis to $D_{\text{edge}} = 0.40 \text{ m}^2/\text{s}$ at the edge, which is representative of moderate anomalous transport. The pinch velocity profiles for B^{1+} through B^{5+} follow Eq. (9) and peak in the outer half of the plasma ($\rho \approx 0.6$ to 0.8). The neutral B^0 receives no pinch (since a neutral atom is unaffected by the radial electric field), and B^{5+} receives the full amplitude of -0.08 m/s .

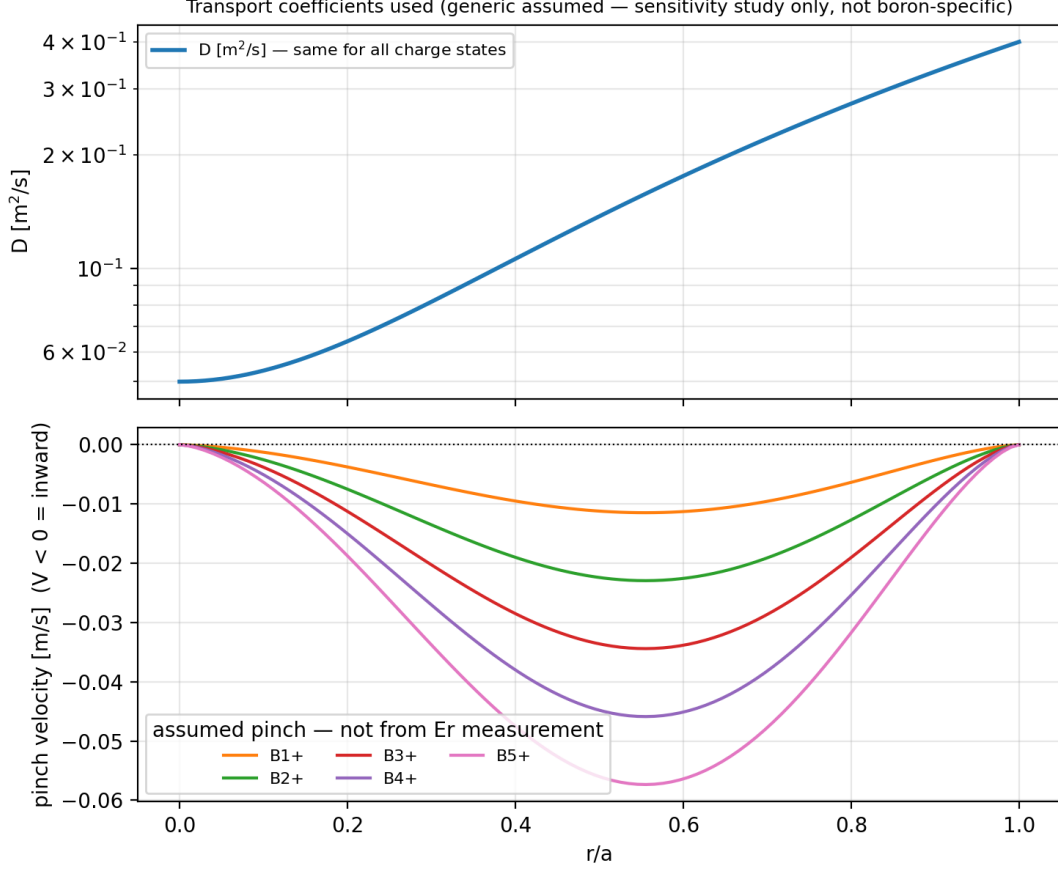


Figure 3: (03_transport_coefficients_used.png) Transport coefficients for the medium-D, charge-scaled-pinch reference case. *Top:* Generic anomalous diffusivity profile $D(\rho)$, which is the same for all charge states in this run. *Bottom:* Assumed inward pinch velocity $V_Z(\rho)$ for B^{1+} through B^{5+} (color-coded), with $V_{B^0} = 0$. These profiles are not derived from a measurement or neoclassical calculation.

4.4 Damköhler Timescale Map (Figure 4)

A useful way to characterize the transport to ionization competition for neutral boron is through the Damköhler number,

$$\text{Da} \equiv \frac{\tau_D}{\tau_{\text{iz}}}, \quad \tau_D = \frac{L^2}{D}, \quad \tau_{\text{iz}} = \frac{1}{S_{B^0}^{\text{ion}}(r)}, \quad (10)$$

where L is the source deposition width ($L = \sigma_{\text{source}} \times a \approx 4.2 \text{ mm}$). $\text{Da} > 1$ means ionization is faster than diffusion over the source footprint, so neutrals are ionized locally before spreading radially. $\text{Da} < 1$ means diffusion is faster, and neutrals can spread before ionizing.

Figure 4 shows $\log_{10}(\text{Da})$ as a 2D map in $(r/a, D)$ space (04_damkohler_timescale_map.png). Across the full range of D scanned (10^{-4} to $\sim 3 \text{ m}^2/\text{s}$), the Damköhler number remains above unity throughout most of the plasma, confirming that neutral boron is ionized very close to its deposition location. The white line shows the D profile used in the reference run, which sits comfortably in the $\text{Da} > 1$ regime. The conclusion is that the detailed neutral source shape is not the dominant factor for core penetration, what matters is the transport of the resulting ions.

It is worth noting that L here is a fixed global parameter (the source deposition width), not a local gradient scale length. The Damköhler map is therefore a statement about the source geometry, not a local transport regime analysis.

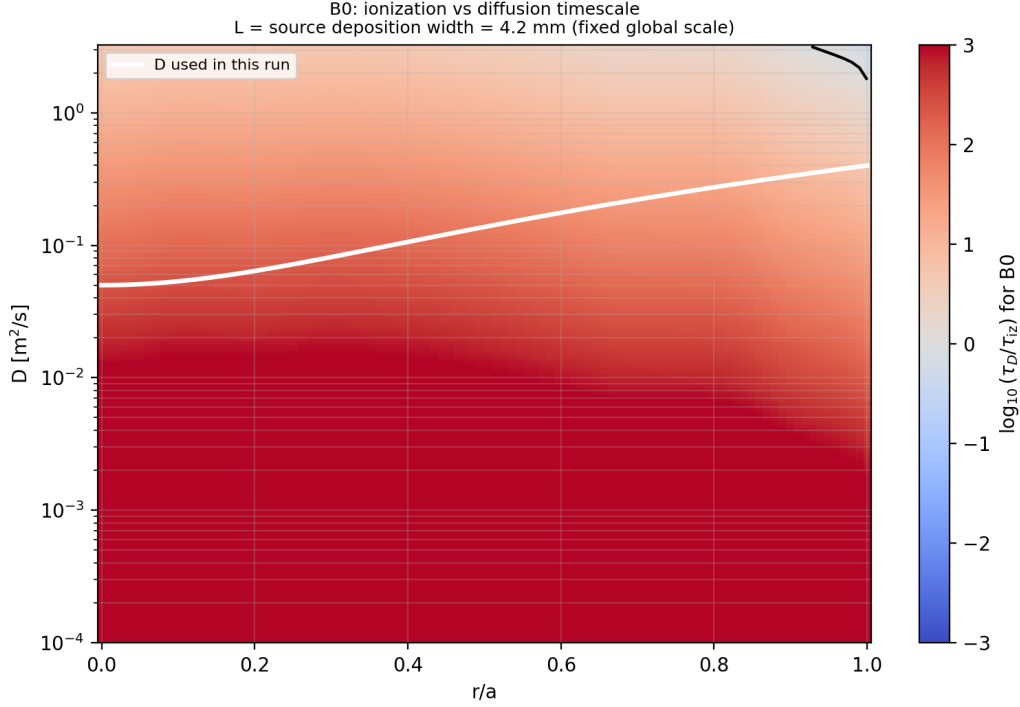


Figure 4: (04_damkohler_timescale_map.png) $\log_{10}(\tau_D/\tau_{iz})$ for neutral boron as a function of radius and diffusivity D . Red ($Da > 1$): ionization faster than diffusion over the source width $L = 4.2$ mm. Blue ($Da < 1$): diffusion faster. The black contour marks $Da = 1$. The white line shows the D profile used in the reference simulation. Over the entire scan range, neutral boron is ionization-dominated, confirming that core penetration is controlled by ion transport, not neutral penetration.

4.5 Charge-State Profiles and Coronal Equilibrium (Figure 5)

Figure 5 shows the final boron charge-state profiles (05_charge_states_final.png). The left panel shows the spatial shapes of each charge state’s density, normalized to the overall maximum, with the fractional population at $\rho = 0.5$ annotated. In the reference (medium D) case, B^{5+} dominates the core, as expected given the high core temperature ($T_e \sim 900$ eV). Lower charge states peak progressively further out, forming a nested shell structure characteristic of impurity transport in a hot plasma.

The right panel compares the time-dependent simulation charge-state fractions (solid) with the local coronal equilibrium fractions computed from Eq. (5) (dashed). In the core ($\rho < 0.4$), the simulation and coronal equilibrium fractions agree reasonably well, indicating that the atomic physics has converged and that transport is not strongly distorting the ionization balance there. In the outer region ($\rho > 0.6$), the simulation deviates from coronal equilibrium, which is expected: the shorter timescales for lower charge states and the presence of the source near $\rho = 0.98$ mean that the edge is not in equilibrium. The overall shape of the charge-state distribution is physically reasonable.

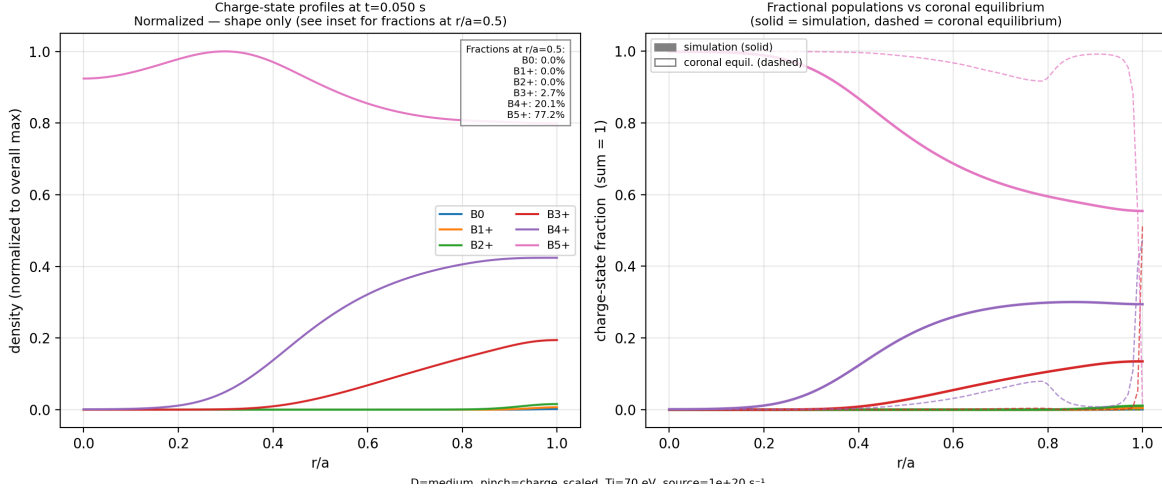


Figure 5: (05_charge_states_final.png) Final boron charge-state profiles. *Left:* Density of each charge state B⁰ through B⁵⁺, normalized to the overall maximum. The inset shows fractional populations at $\rho = 0.5$. *Right:* Charge-state fractions from the time-dependent simulation (solid) compared to local coronal equilibrium (dashed). Good agreement in the core confirms atomic physics self-consistency; deviations near the edge reflect the proximity of the source.

4.6 Core Penetration and Peaking Factor (Figure 6)

Figure 6 shows two metrics for core penetration as a function of time (06_core_fraction_vs_time.png). The core fraction is the radially weighted particle inventory inside $\rho < 0.5$ divided by the total inventory, this is a volume fraction that accounts for the cylindrical geometry. The peaking factor $\eta \equiv n_{\text{total}}(r = 0) / \langle n_{\text{total}} \rangle_{\text{vol}}$ measures the degree of on-axis concentration; $\eta > 1$ means boron is accumulating on axis.

Both metrics evolve on timescales of ~ 10 to 20 ms, reflecting the transport time across the minor radius ($\tau_{\text{tr}} \sim a^2/D \sim 0.04/0.05 \approx 0.8$ s for the core D , but the effective time to approach steady state is shorter because the source is at the edge and the inner region fills up diffusively). In the medium-D, charge-scaled-pinch case, the core fraction rises from zero and saturates at a level that depends on the transport coefficients, the main result explored in the scan below.

4.7 Sensitivity to Anomalous Diffusivity (Figures 7 and 8)

The transport scan over low, medium, and high diffusivity cases is the central result of the project. Figure 7 shows the three $D(\rho)$ profiles (01_diffusivity_cases.png from the scan output), spanning nearly two orders of magnitude in core diffusivity (0.01 to $0.15 \text{ m}^2/\text{s}$).

Figure 8 shows the core fraction time evolution for the three cases (02_core_fraction_scan.png) and the final core fraction bar chart (05_scan_summary_bar.png). The result is qualitatively clear: lower diffusivity leads to higher core penetration. This makes sense physically, slower transport allows the inward pinch to accumulate boron on axis before diffusion can spread it back out. In the low-D case, the time-to-saturation is also longer because the timescale to fill the core scales as a^2/D .

Quantitatively, the spread in core fraction across the three cases demonstrates that the answer is genuinely uncertain at this stage: without knowing the true anomalous diffusivity in HSX (which would require fitting to boron spectroscopy data or a turbulence calculation), a factor of 10 to 100

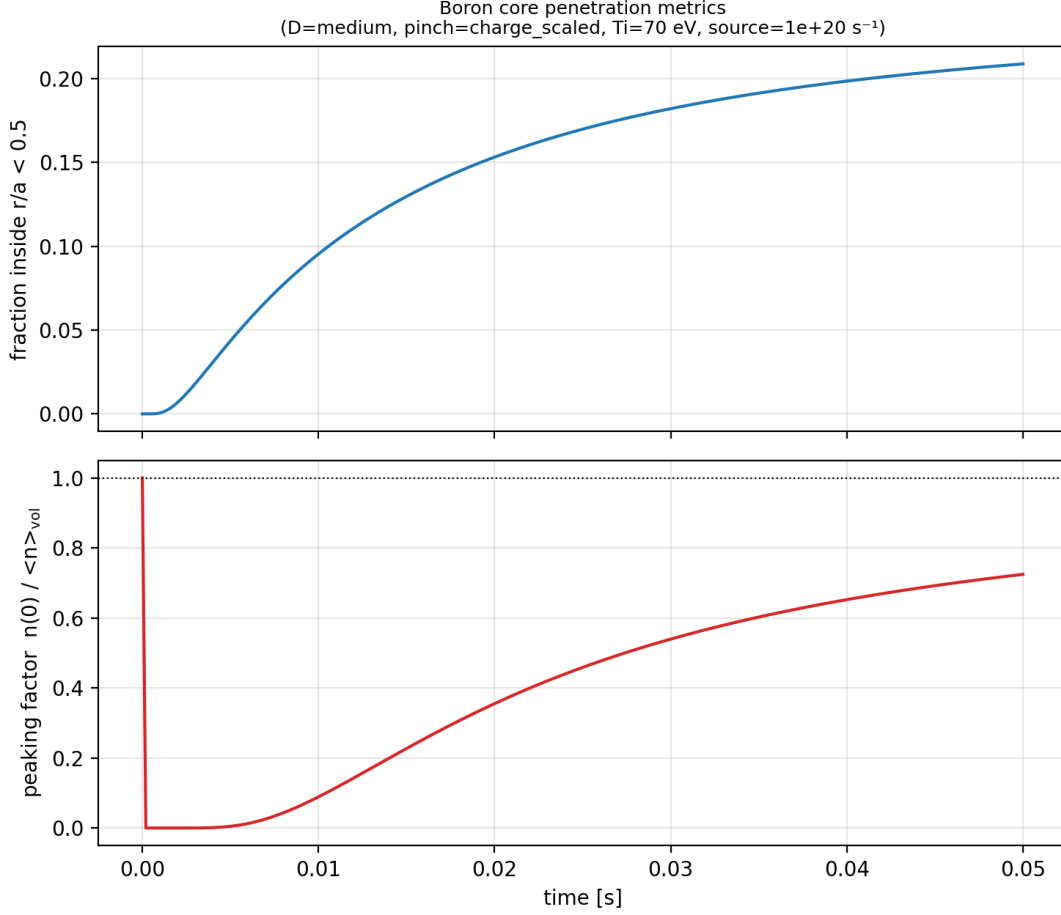


Figure 6: (06_core_fraction_vs_time.png) Boron core penetration metrics as a function of time for the reference case (medium D , charge-scaled pinch, $T_i = 70$ eV). *Top:* Core fraction, defined as the volume-weighted boron inventory inside $\rho < 0.5$ divided by the total. *Bottom:* Peaking factor $\eta = n(0)/\langle n \rangle$; values above the dashed line ($\eta = 1$) indicate on-axis accumulation.

uncertainty in core penetration remains. This is the central result of the sensitivity study.

The charge-state scan (03_charge_state_scan.png) shows that the shell structure described in Section 4.5 is qualitatively robust across all three transport cases, though the degree of penetration of each state shifts with D . The peaking factor scan (04_peaking_factor_scan.png) shows that the on-axis concentration is most pronounced in the low- D case, consistent with the inward pinch having more time to build up a peaked profile before diffusion smears it.

5 Discussion

5.1 What the Model Gets Right

Several aspects of the model are physically well-grounded. The finite-volume transport operator correctly implements the cylindrical divergence in Eq. (1), with the toroidal $4\pi^2 R$ factor canceling identically. The ionization/recombination coupling between charge states is exactly particle-conservative. The stiff BDF solver is the appropriate choice given the $\sim 10^4$ ratio between

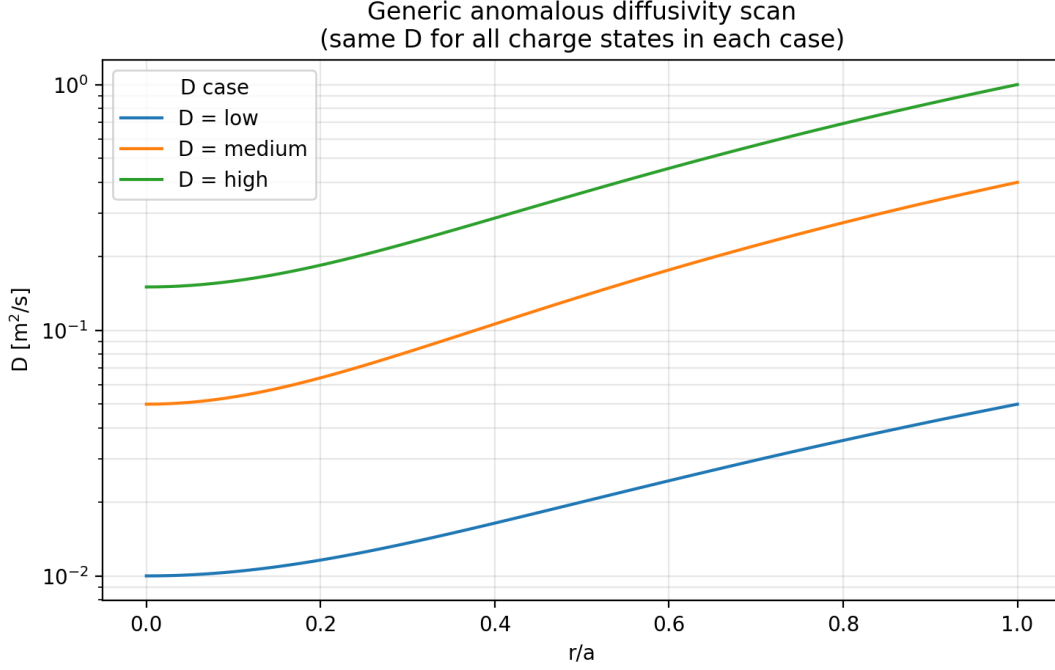
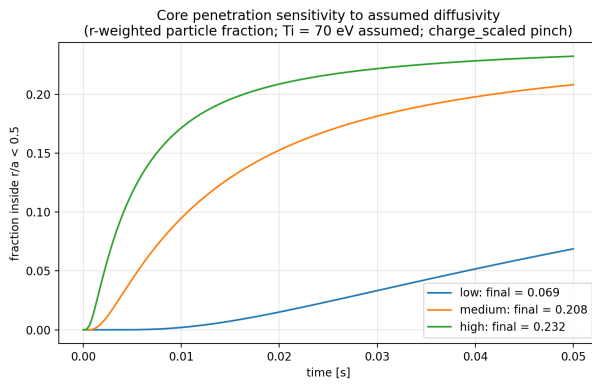
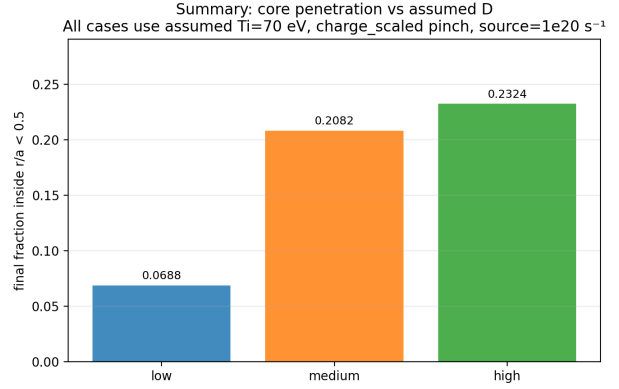


Figure 7: (01_diffusivity_cases.png) Three generic anomalous diffusivity profiles used in the sensitivity scan. All profiles follow $D(\rho) = D_{\text{core}} + (D_{\text{edge}} - D_{\text{core}})\rho^2$. Parameter values are given in Table 1.



(a) Core fraction time evolution.



(b) Final core fraction comparison.

Figure 8: (02_core_fraction_scan.png, 05_scan_summary_bar.png) Core penetration sensitivity to assumed diffusivity. *Left:* Core fraction vs. time for all three D cases. Lower diffusivity results in higher and faster-saturating core penetration. *Right:* Bar chart of final core fraction, showing the spread across cases. The charge-scaled pinch ($T_i = 70$ eV assumed) is held fixed across all three cases.

the fastest ionization timescale and the simulation duration. The coronal equilibrium comparison in Figure 5 provides an internal consistency check that is not commonly shown in simple transport models, and the fact that the simulation approaches coronal equilibrium in the core is a useful sanity check.

The Damköhler analysis in Figure 4 gives an important result: it establishes that for the HSX

temperature and density conditions of shot 37, neutral boron ionizes within microseconds at essentially any location in the plasma. The question of whether boron reaches the core is therefore a question about ion transport, not neutral penetration, which is an important conceptual clarification.

5.2 Limitations and Assumptions

The model has several significant limitations that must be acknowledged before any quantitative conclusions are drawn.

Transport is not boron-specific. The D and V profiles are generic sensitivity inputs, not neoclassical calculations or experimental fits. Boron neoclassical transport in HSX could in principle be computed from SFINCS/PENTA using the VMEC equilibrium, and this would replace the assumed profiles with a physically motivated prediction.

No charge-exchange recombination. CX recombination can be a significant recombination channel in plasmas with substantial neutral hydrogen, and it depends on T_i . It has not been included here. For HSX with relatively low neutral density and moderate T_i , this is probably a second-order effect, but it deserves verification.

Constant T_i . The flat $T_i = 70$ eV profile is a rough estimate. CHERS measurements, when available, would provide a measured $T_i(r)$ profile and would also enable the Er estimate needed for a self-consistent pinch.

The source rate is uncalibrated. The 10^{20} s^{-1} injection rate is an order-of-magnitude placeholder. Because the transport equation is linear in density, the source rate affects the absolute boron density but not the charge-state fractions or core fraction, so the shape of the results is insensitive to this number, but any comparison to an absolute measurement (e.g., a B emission brightness) would require calibration.

Cylindrical geometry. The 1D cylindrical approximation ignores flux surface shaping and the actual 3D structure of HSX. A full neoclassical calculation in HSX would need to account for the helical geometry.

5.3 Connection to Course Concepts

This project connects directly to several topics from the course.

The transport equation (Eq. 1) is a direct application of the radial particle flux treatment from the transport chapter, including the role of diffusion and convection as separate physics channels. The charge-state coupling through ionization and recombination is an application of atomic physics in a plasma context, specifically the role of the ionization balance in setting the charge distribution.

The Damköhler analysis is a dimensionless-number approach to characterizing which physical process dominates in a given regime, which is a useful way to organize plasma physics problems. The identification of the $\sim 2 \mu\text{s}$ neutral ionization timescale versus the ~ 10 to 50 ms transport timescale is the kind of order-of-magnitude analysis that sits at the heart of plasma physics intuition.

The coronal equilibrium concept (Section 2.3) connects to atomic physics and the idea that in many fusion plasmas, especially in the core, the charge-state distribution is close to the local equilibrium set by ionization and recombination rates. Deviations from coronal equilibrium are a diagnostic of transport effects, which is a central concept in impurity transport theory.

Finally, the peaking factor and core fraction metrics connect to the broader question of impurity accumulation in fusion devices, which is a central concern in magnetic confinement fusion research because peaked impurity profiles can lead to radiation collapse.

6 Conclusions and Future Work

This project built a standalone 1D radial impurity transport model for boron in HSX and used it to explore how sensitive the core penetration of injected boron powder is to uncertain transport parameters. The main findings are:

1. Neutral boron ionizes extremely rapidly ($\tau \sim 1$ to $2\mu\text{s}$) anywhere in the HSX plasma under shot-37 conditions. Core penetration is therefore controlled by ion transport, not neutral penetration, and the Damköhler number is large ($\gg 1$) for all reasonable diffusivities.
2. The charge-state distribution in the core approaches local coronal equilibrium, confirming internal consistency of the atomic physics model. B^{5+} dominates the core, and lower charge states form a nested shell structure decreasing toward the edge.
3. Core penetration (quantified by both the core fraction inside $\rho < 0.5$ and the peaking factor) is sensitive to the assumed anomalous diffusivity by roughly an order of magnitude across the range scanned. Lower diffusivity, combined with an inward pinch, leads to substantially higher core accumulation.
4. The model gives a useful starting point and can be extended. The main things needed to make it predictive are: (a) boron-specific neoclassical D and V from SFINCS/PENTA; (b) CHERS-measured $T_i(r)$ and $E_r(r)$; and (c) a source rate calibrated from spectroscopy or dropper measurements.

One thing I learned from this project is that the Damköhler analysis changed how I thought about the problem. Before working through the numbers, I expected that the question of whether boron reaches the core might depend on neutral penetration depth. Instead, the calculation showed clearly that neutral boron ionizes on timescales so short that neutral transport is essentially irrelevant, the relevant physics is entirely in the ion transport. That shift in intuition was the most valuable outcome of the modeling.

Reproducibility

All code is available in the project repository. To reproduce the main results:

```
pip install numpy scipy matplotlib
python run_model.py to ti 70 to D-case medium to pinch charge_scaled
python run_transport_scan.py
```

Outputs are written to `OUTPUT/<run_tag>/`. A complete description of each plot and all input assumptions is given in `README.md`. The `metadata.json` file in each output directory records all run parameters and distinguishes measured from assumed inputs.

Required data files (not included in the repository due to size): `DATA/2026-04-15_shot37_t0p83-0p85.mat` (HSX Thomson data), `DATA/scd89_b.dat` and `DATA/acd89_b.dat` (OpenADAS boron rate files).